QUESTION PAPER

Model Paper — 02

Mathematics II

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Time: 3 hours (+ 10 minutes reading time) **Total marks:** 100

Index number:

Instructions:

- Use the additional reading time of 10 minutes to read the paper, to select the questions you will answer and to decide the order in which you will answer them.
- Select **five** questions from **Part A** and **five** questions from **Part B**, and answer **ten** questions in all. Write your answers on separate answer sheets.
- Show the relevant steps and write the correct units in your answers.
- Each question carries **10 marks**. Where a question asks you to copy a figure, copy it on to your answer sheet.
- In constructions use only a straight edge with a cm/mm scale and a pair of compasses; the construction lines must be clearly visible.
- The volume of a right pyramid of base area A and height h is $\frac{1}{3}Ah$. The volume of a right circular cylinder of base radius r and height h is $\pi r^2 h$.

Part A

Answer only five questions.

1. Amaya and Pradeep each want to buy a photocopier for a library. Its cash price is Rs. 60 000.

Amaya's method	Pradeep's method
She takes a bank loan of Rs. 60 000 for one year at a simple interest rate of 14% per annum, and settles the loan with the interest in one payment at the end of the year.	He pays the price in 12 equal monthly instalments. Each instalment includes Rs. 5 000 of the price, and the total interest charged for the year is Rs. 5 460. The interest is charged on the part of the price that remains unpaid each month (the reducing balance).

(a) Find the amount Amaya pays to settle her loan. (2 marks)

(b) Find the amount of one monthly instalment that Pradeep pays. (2 marks)

(c) Show that the annual interest rate under Pradeep's method is higher than the annual interest rate under Amaya's method. (6 marks)

2. The incomplete table below gives values of the function $y = x^2 - 4x - 5$ for $-2 \leq x \leq 6$. The graph of the function is symmetric about a vertical line.

x	−2	−1	0	1	2	3	4	5	6
y	7	0	−5	−8	−9	−8	....	0	7

(i) Using the symmetry of the graph, write the value of y when $x = 4$. (1 mark)

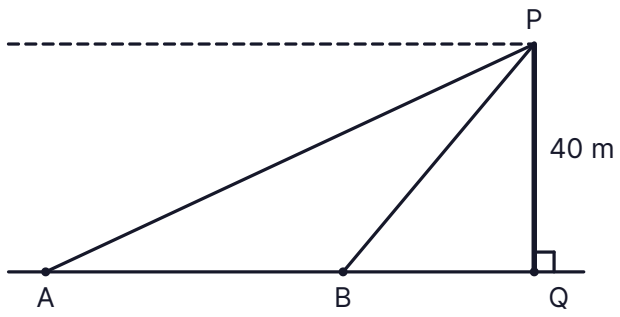
(ii) Using a suitable scale on both axes, draw the graph of the function on graph paper. (3 marks)

(iii) (a) Write the values of x at the points where the graph cuts the x-axis. (1 + 1 marks)
(b) Hence write the function in the form $y = (x + p)(x + q)$.

(iv) The straight line $y = -6$ meets the graph at the points A and B. (2 + 2 marks)
(a) Using the graph, find the range of values of x for which $y < -6$.
(b) Find the length of AB, correct to the first decimal place.

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3. PQ is the vertical chimney of a factory, standing on level ground, and PQ = 40 m. A bus is at the point A and a car is at the point B on the ground, as shown in the figure. The points A, B and Q are in a straight line. From P the angle of depression of A is 25° and the angle of depression of B is 50°. Copy the figure on to your answer sheet.



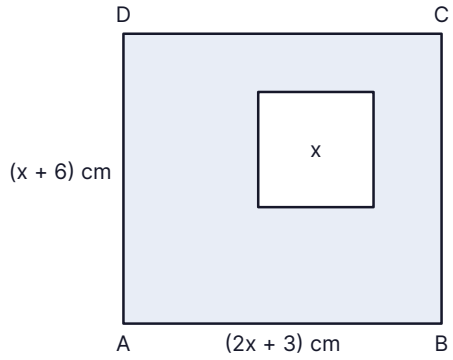
(i) On your figure mark the two angles of depression, and write the size of PÂQ and of PB̂Q with the reason. (2 marks)

(ii) Using tables, find the distances AQ and BQ, correct to two decimal places. (4 marks)

(iii) Show that the distance between the bus and the car is greater than 52 m. (4 marks)

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4. In a factory a rectangular metal sheet ABCD with AB = $(2x + 3)$ cm and BC = $(x + 6)$ cm has a square hole of side x cm cut out of it, as shown in the figure. The area of the metal that remains is 93 cm^2.



(i) Show that $x^2 + 15x - 75 = 0$. (3 marks)

(ii) Solve the equation and find the value of x correct to the nearest centimetre. (Take $\sqrt{21} = 4.58$.) (3 marks)

(iii) Show that the perimeter of the rectangle ABCD is $(15\sqrt{21} - 27)$ cm and hence show that it is greater than 40 cm. (4 marks)

5. A bus takes a group of 40 people, adults and children, on a trip. An adult ticket costs Rs. 90 and a child ticket costs Rs. 60. The total money collected from the tickets was Rs. 2 850. Let the number of adults be x and the number of children be y.

(i) Form a pair of simultaneous equations in x and y, and solve them to find the number of adults and the number of children. (5 marks)

(ii) Each adult carried the same number of books to read on the trip, and each child carried 2 books fewer than each adult. The group had 190 books altogether. Find the number of books carried by one adult. (5 marks)

6. The table shows the number of books borrowed from a library on each of 30 days.

Number of books borrowed	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69
Number of days	1	2	6	8	7	6

(i) Find the mean number of books borrowed per day. (3 marks)

(ii) Using this mean, find the number of books that can be expected to be borrowed in 24 days. (2 marks)

(iii) The library earns Rs. 5 for each book borrowed. Find the income that can be expected in 24 days. (3 marks)

(iv) Find the minimum possible total number of books borrowed on the days on which fewer than 30 books were borrowed. (2 marks)

Part B

Answer only five questions.

7. Use only a straight edge with a cm/mm scale and a pair of compasses. The construction lines must be clearly visible.

(i) Construct the line segment AB = 9 cm. At A construct an angle of 60° with AB, and bisect it. (3 marks)

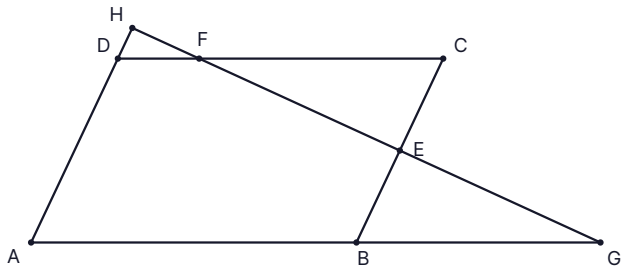
(ii) Mark the point C on the bisector such that AC = AB, and join BC. (2 marks)

(iii) Construct the perpendicular bisector of AC, and with AC as the diameter construct a semicircle on the side of AC opposite to B. (2 marks)

(iv) Through A construct the line parallel to BC. Name as D the point where it meets the semicircle, and join CD. (2 marks)

(v) Write down the size of $A\hat{C}D$, with reasons. (1 mark)

8. In the parallelogram ABCD, E is the mid-point of BC and F is a point on DC such that DF = $\frac{1}{4}$ DC. AB produced and FE produced meet at G, and AD produced and EF produced meet at H. Copy the figure on to your answer sheet.



(i) Show that the triangles BGE and CFE are congruent. (3 marks)

(ii) Join BF and GC. Show that BGCF is a parallelogram. (2 marks)

(iii) Show that the triangles DFH and BGE are equiangular. (2 marks)

(iv) Hence show that DH = $\frac{1}{6}$ AD. (3 marks)

9. A solid metal cylinder of base radius r and height $2r$ is melted, without any loss of metal, and recast into 6 identical solid pyramids. Each pyramid has a square base of side a and a height of $3a$.

(i) Show that $a^3 = \frac{\pi}{3} r^3$. (4 marks)

(ii) Taking $r = 8.1$ and $\pi = 3.14$, find the value of a^3 using logarithm tables, and hence find a correct to the nearest centimetre. (3 marks)

(iii) A pyramid of base side a and height $3a$ is cut out of a solid metal cuboid with a square base of side a and a height of $3a$. Using the value of a found in (ii), find the volume of the metal that remains. (3 marks)

10. Sanduni and Amaya each have to arrange the same collection of books on the shelves of a library. Sanduni arranges 15 books on the first day and, on each following day, 3 books more than on the previous day. She completes the work on the 16th day.

(i) Write the number of books Sanduni arranges on each of the first three days. (1 mark)

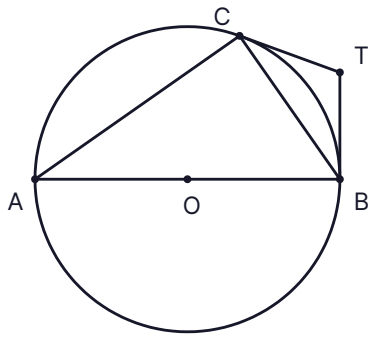
(ii) Find the number of books she arranges on the 16th day. (2 marks)

(iii) Find the total number of books in the collection. (2 marks)

(iv) Amaya arranges 2 books more each day than on the previous day, and completes the same work on the 15th day. Find the number of books she arranges on the first day. (3 marks)

(v) On which day do Sanduni and Amaya arrange the same number of books? (2 marks)

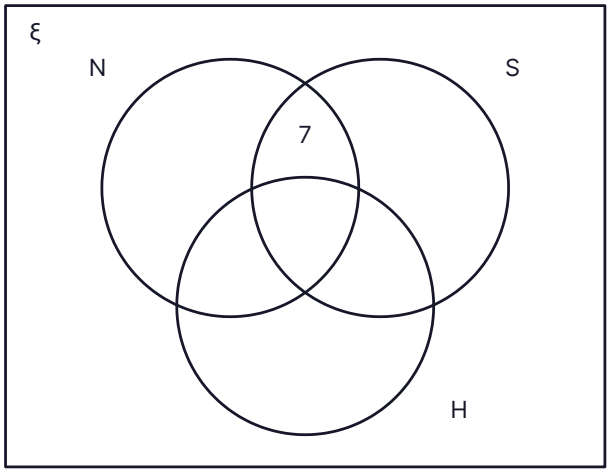
11. In the figure, O is the centre of the circle and AB is a diameter. The tangents drawn to the circle at B and at C meet at T. Copy the figure on to your answer sheet and join OC and OT.



(i) (a) Show that OBTC is a cyclic quadrilateral. (2 + 4 marks)
(b) Show that OT is parallel to AC.

(ii) Name two angles that are equal to $T\hat{B}C$, giving a reason for each. (4 marks)

12. There are 82 members in a library. Each of them may borrow Novels, Science books and History books. The sets N, S and H are the sets of members who borrowed Novels, Science books and History books respectively. 10 members borrowed none of the three types, 40 members borrowed History books, and 32 members borrowed two or more of the types. 7 members borrowed only Novels and Science books. Copy the Venn diagram on to your answer sheet and fill in the information.



(i) Find the number of members who borrowed exactly one of the three types. (3 marks)

(ii) The number of members who borrowed all three types is equal to the number who borrowed exactly two types. Find the number who borrowed all three types. (3 marks)

(iii) The number who borrowed only Science and History books is twice the number who borrowed only Novels and History books. Find the number who borrowed only Novels and History books, and the number who borrowed only History books. (2 marks)

(iv) If 38 members borrowed Novels, find the number of members who borrowed only Science books. (2 marks)

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